Introduction:

The North Chennai Thermal Power Plant Station—II is a subcritical plant run by TANGEDCO under the TNEB, comprising two units, each with a capacity of 600 MW. It has been functioning since 2014 on 10,000 acres of land in Attipattu Pudhunagar, Chennai. This thermal power plant's feed is bituminous coal with a calorific value of 3100-3500 kcal/kg and a fine grinding size of 20 to 25 mm. Mixing two distinct types of coals improves plant efficiency while maintaining calorific content. One of the two coals comes from Odisha, Chhattisgarh, and other parts of India, while the other is imported from Indonesia and America and has a calorific content of about 5000 kcal/kg. The factory won national praise for its excellent productivity, low auxiliary power use, and fuel economy. Belt conveyors are used to transport coal securely from the yard to the plant, minimizing spillage. This plant uses pollution control equipment, including electrostatic precipitators, to reduce particle emissions and processes wastewater before it is discharged into the sea. It has experienced several problems, yet it plays a crucial role in meeting the electricity demands of Chennai and other regions of the state through its robust infrastructure.

Fig 1.1 Thermal station

2. OVERVIEW OF THERMAL POWER PLANT

Introduction

A thermal power plant is a generating station that converts the chemical energy stored in fuel into electrical energy. In India, coal-based thermal power plants play a significant role in meeting the country's growing electricity demand. These plants generate power by burning coal to produce high-pressure steam, which drives turbines connected to electrical generators. The generated electricity is then transmitted to consumers through transmission and distribution networks.

North Chennai Thermal Power Station (NCTPS) Stage-II is one of the major thermal power plants operated by Tamil Nadu Power Generation and Distribution Corporation (TANGEDCO). The plant contributes significantly to the electricity requirements of Tamil Nadu by ensuring reliable and continuous power generation. The station utilizes modern equipment and advanced control systems to achieve efficient and safe operation.

Principle of Thermal Power Generation

The working principle of a thermal power plant is based on the Rankine Cycle. In this process, coal is burned in a boiler to generate heat energy. The heat converts water into high-pressure and high-temperature steam. This steam is directed onto turbine blades, causing the turbine shaft to rotate at high speed. The turbine is mechanically coupled with a generator, which converts the mechanical energy into electrical energy through electromagnetic induction.

After passing through the turbine, the steam loses its energy and enters a condenser, where it is cooled and converted back into water. The condensed water is then pumped back to the boiler, completing the cycle. This continuous process enables uninterrupted electricity generation.

Coal Handling System

Coal is the primary fuel used in thermal power plants. At NCTPS Stage-II, coal is received from various sources through railway wagons and ships. The coal is unloaded, stored in coal yards, and transported through conveyor belts to the crusher house. Large coal particles are crushed into smaller sizes and then stored in bunkers.

From the bunkers, coal is fed into pulverizing mills where it is converted into fine powder. The pulverized coal is mixed with air and supplied to the boiler furnace for efficient combustion. Proper coal handling is essential to ensure a continuous fuel supply and stable plant operation.

Boiler System

The boiler is one of the most important components of a thermal power plant. It converts water into high-pressure steam using the heat generated from coal combustion. Pulverized coal is burned inside the furnace, producing temperatures above 1300°C. Water circulating through tubes absorbs this heat and transforms into steam.

The boiler consists of several important components such as the furnace, water walls, boiler drum, superheater, reheater, economizer, and air preheater. These components improve heat transfer efficiency and ensure optimum steam generation. The superheater increases the steam temperature before it enters the turbine, while the reheater reheats partially expanded steam to improve cycle efficiency.

Turbine System

The steam turbine converts the thermal energy of steam into mechanical energy. High-pressure steam enters the High-Pressure (HP) turbine and expands through multiple stages. The steam is then reheated and directed to the Intermediate-Pressure (IP) turbine, followed by the Low-Pressure (LP) turbines.

The turbine shaft rotates at approximately 3000 RPM, matching the 50 Hz frequency requirement of the power system. Proper lubrication, cooling, and governing systems are necessary to ensure reliable turbine operation. Regular maintenance activities such as vibration monitoring, alignment checks, and bearing inspections help prevent failures and improve equipment life.

Generator System

The generator is coupled directly to the turbine shaft and converts mechanical energy into electrical energy. It operates on the principle of electromagnetic induction. When the rotor rotates inside the stator windings, a changing magnetic field is produced, inducing an electromotive force (EMF) in the stator conductors.

The electrical power generated at approximately 21 kV is transmitted to transformers, where the voltage is increased to suitable transmission levels. This generated power is then supplied to the state electrical grid for distribution to residential, commercial, and industrial consumers.

Condenser and Feed Water System

After performing work in the turbine, the exhaust steam enters the condenser. Cooling water absorbs heat from the steam, converting it back into water. The condensate is collected in hot wells and pumped through feedwater heaters and deaerators before being sent back to the boiler.

The feedwater system improves thermal efficiency by preheating the water before it enters the boiler. This process reduces fuel consumption and increases overall plant performance. Maintaining high-quality feedwater is essential to prevent corrosion, scaling, and damage to boiler tubes.

Cooling Water System

A large quantity of cooling water is required to condense the exhaust steam from the turbine. NCTPS Stage-II uses seawater as the primary cooling medium. The water is drawn from the sea, circulated through condensers, and discharged after appropriate treatment.

The cooling water system plays a vital role in maintaining condenser vacuum and improving turbine efficiency. Continuous monitoring and treatment of cooling water help prevent fouling, corrosion, and environmental pollution.

Ash Handling System

Coal combustion produces large quantities of fly ash and bottom ash. Fly ash is carried with flue gases and is collected using Electrostatic Precipitators (ESPs). Bottom ash settles at the bottom of the boiler furnace and is removed through hydraulic systems.

The collected ash is transported to storage silos or ash ponds for disposal and reuse. Fly ash is commonly utilized in cement manufacturing, brick production, and road construction projects. Effective ash handling systems help reduce environmental pollution and ensure compliance with environmental regulations.

Instrumentation and Control System

Modern thermal power plants utilize advanced instrumentation and control systems to monitor and regulate plant operations. Parameters such as pressure, temperature, flow rate, water level, and steam conditions are continuously measured using sensors and transmitters.

Distributed Control Systems (DCS) and Supervisory Control and Data Acquisition (SCADA) systems provide centralized monitoring and automatic control of plant equipment. These systems improve operational efficiency, safety, and reliability while minimizing human intervention.

Environmental Protection Measures

Environmental protection is a critical aspect of thermal power plant operation. Various pollution control systems are employed to minimize the impact of plant emissions. Electrostatic Precipitators remove fly ash from flue gases, while proper ash disposal methods reduce land and water pollution.

Water treatment systems ensure that discharged water meets environmental standards. Continuous monitoring of emissions and adherence to environmental regulations help maintain sustainable plant operation

3. MAIN CHEMISTRY LABORATORY (DCC)

Introduction

The Main Chemistry Laboratory, also known as the Deputy Chief Chemist (DCC) Department, plays a vital role in the efficient and reliable operation of North Chennai Thermal Power Station (NCTPS) Stage-II. The primary responsibility of this department is to maintain the chemical quality of water, steam, cooling water, and fuel used in the power generation process. Proper chemical monitoring and treatment help prevent corrosion, scaling, and deposition in boilers, turbines, condensers, and other plant equipment. The department ensures that all chemical parameters remain within permissible limits, thereby improving plant efficiency and extending equipment life.

Objectives of the Chemistry Laboratory

The main objectives of the Chemistry Laboratory are:

- To ensure the supply of high-purity demineralized water for boiler operation.
- To monitor the quality of feedwater, condensate, and cooling water.
- To analyze coal quality for efficient combustion.
- To prevent corrosion and scale formation in plant equipment.
- To ensure compliance with environmental regulations regarding water discharge and emissions.

These activities are essential for maintaining safe and efficient power plant operation.

RO-DM Plant

One of the major sections under the Chemistry Department is the Reverse Osmosis–Demineralization (RO-DM) Plant. This plant is responsible for producing high-purity water required for boiler feed applications. Raw water undergoes several stages of treatment, including filtration, ultrafiltration, reverse osmosis, and demineralization.

The treatment process removes suspended solids, dissolved salts, minerals, and other impurities from water. The final demineralized water has very low conductivity and is suitable for use in high-pressure boilers. This helps prevent scaling, corrosion, and efficiency losses in the steam generation system.

Water Testing Laboratory

The Water Testing Laboratory continuously monitors the chemical quality of water used throughout the power plant. Samples are collected from various locations, including feedwater systems, condensate systems, cooling water circuits, and wastewater treatment facilities.

Several important parameters are analyzed regularly, such as:

- pH Value
- Conductivity
- Silica Content
- Dissolved Oxygen
- Hardness
- Phosphate Content

- Chlorides
- Iron Concentration

Maintaining these parameters within specified limits is essential for preventing corrosion, scale formation, and equipment damage. Regular testing also ensures environmental compliance and reliable plant operation.

Coal Analysis Laboratory

The Coal Analysis Laboratory evaluates the quality of coal used for power generation. Since coal is the primary fuel for thermal power plants, its quality directly affects combustion efficiency, boiler performance, and emissions.

The laboratory analyzes various coal properties, including:

- Moisture Content
- Ash Content
- Volatile Matter
- Fixed Carbon
- Gross Calorific Value (GCV)

A Bomb Calorimeter is used to determine the calorific value of coal. The measured heat energy helps determine the fuel's efficiency and suitability for power generation. Regular coal testing ensures optimum combustion and reduced operational costs.

Cooling Water Chemistry

The Chemistry Department also supervises the cooling water treatment process. Seawater used in condensers and cooling systems contains biological organisms and dissolved impurities that can cause fouling and corrosion.

To prevent these issues, chemicals such as sodium hypochlorite and other biocides are added to control biological growth. The cooling water is continuously monitored and treated before discharge to minimize environmental impact. Proper cooling water management helps maintain condenser efficiency and ensures safe operation of the plant.

Importance of the Chemistry Laboratory

The Chemistry Laboratory serves as the backbone of water and fuel quality management in the thermal power plant. Through continuous monitoring, testing, and treatment, the department helps improve efficiency, reduce maintenance costs, prevent equipment failures, and ensure environmental safety. Without effective chemical control, the reliability and performance of the power plant would be significantly affected.

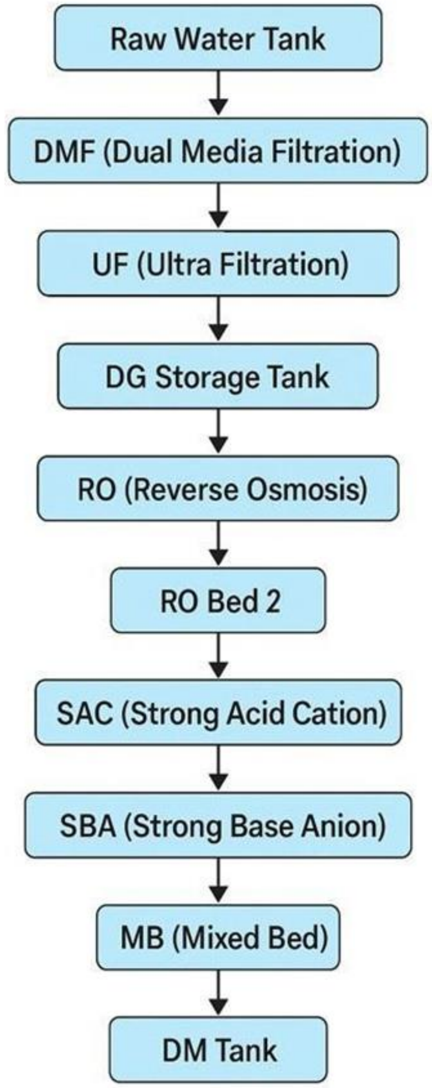

Fig 3.1 Water Treatment Process Flowchart

4. ELECTRICAL MAINTENANCE (EM)

Introduction

The Electrical Maintenance (EM) Department at North Chennai Thermal Power Station Stage-II (NCTPS-II) plays a crucial role in ensuring the reliable and continuous operation of the power plant. The department is responsible for the maintenance, testing, inspection, and protection of electrical equipment used in power generation and distribution. Electrical systems form the backbone of the plant, as they support the operation of generators, transformers, motors, switchgear, control systems, and auxiliary equipment. Proper maintenance practices help prevent equipment failures, reduce downtime, improve operational efficiency, and ensure safe working conditions.

Generator and Transformer Maintenance

Generators are the primary electrical machines responsible for converting mechanical energy from the turbine into electrical energy. At NCTPS Stage-II, continuous monitoring and maintenance of generators are essential for uninterrupted power generation. Maintenance activities include insulation resistance testing, temperature monitoring, vibration analysis, and inspection of cooling systems.

Transformers are equally important as they step up the generated voltage for transmission and supply power to various auxiliary systems within the plant. The Electrical Maintenance Department performs transformer oil testing, insulation checks, winding resistance measurements, and cooling system inspections. These activities help prevent insulation failure, overheating, and unexpected breakdowns.

Switchgear and Protection Systems

Switchgear equipment is used to control, protect, and isolate electrical circuits. It includes circuit breakers, isolators, busbars, relays, and control panels. Regular maintenance ensures that these components operate correctly during both normal and fault conditions.

Protection systems are designed to safeguard valuable plant equipment from electrical faults such as short circuits, overloads, and earth faults. Protective relays continuously monitor system parameters and automatically disconnect faulty equipment when abnormal conditions occur. Proper testing and calibration of protection systems improve plant safety and reliability.

Motor Maintenance and Auxiliary Equipment

A large number of electrical motors are installed throughout NCTPS Stage-II to operate pumps, fans, compressors, conveyors, and other auxiliary systems. These motors are essential for supporting boiler operation, cooling systems, coal handling, and water treatment processes.

Maintenance activities include lubrication of bearings, insulation testing, vibration monitoring, alignment checks, and temperature measurement. Regular inspection helps identify potential issues before they result in equipment failure. Proper motor maintenance contributes significantly to efficient plant operation.

Battery Banks and Emergency Power Supply

Battery banks and DC supply systems provide backup power for protection circuits, control systems, emergency lighting, and switchgear operation during power interruptions. The Electrical Maintenance Department regularly checks battery voltage, charging systems, electrolyte levels, and terminal conditions.

Emergency power systems ensure that critical plant operations remain functional even during unexpected electrical failures. These systems play a vital role in maintaining plant safety and preventing operational disruptions.

Preventive Maintenance Practices

Preventive maintenance is an important activity carried out by the Electrical Maintenance Department. Routine inspections, thermal scanning, insulation testing, cable checks, and equipment cleaning are performed according to scheduled maintenance plans. These practices help detect faults at an early stage and reduce the possibility of sudden equipment breakdowns.

Maintenance records are maintained systematically to monitor equipment performance and plan future maintenance activities effectively. This approach improves equipment reliability and reduces maintenance costs.

5. INSTRUMENTATION AND CONTROL (C&I)

Introduction

The Instrumentation and Control (C&I) Department at North Chennai Thermal Power Station Stage-II (NCTPS-II) plays a vital role in ensuring the safe, efficient, and reliable operation of the power plant. The department is responsible for monitoring, measuring, and controlling various process parameters such as temperature, pressure, flow rate, water level, and steam conditions. Modern thermal power plants rely heavily on instrumentation systems and automated control technologies to maintain stable operation and achieve maximum efficiency. The C&I department ensures that all process variables remain within prescribed limits and helps prevent equipment failures and operational disturbances.

Role of the C&I Department

The primary function of the C&I Department is to continuously monitor plant parameters and provide accurate data for operation and control. The department maintains various field instruments, sensors, transmitters, analyzers, and control systems installed throughout the plant. These instruments provide real-time information that enables operators to make informed decisions and maintain safe operating conditions.

The department also supports automation systems that reduce manual intervention and improve operational accuracy. Effective monitoring and control contribute significantly to plant efficiency, equipment reliability, and personnel safety.

Process Instruments Used in NCTPS-II

Various measuring instruments are installed throughout the power plant to monitor critical process parameters.

Pressure Measurement Instruments

Pressure transmitters are used to measure steam pressure, feedwater pressure, air pressure, and other process pressures. These instruments provide continuous feedback to the control system, enabling operators to maintain stable plant conditions.

Temperature Measurement Instruments

Temperature is measured using Resistance Temperature Detectors (RTDs) and Thermocouples. These sensors are installed in boilers, turbines, generators, transformers, and auxiliary systems. Accurate temperature monitoring helps prevent overheating and equipment damage.

Flow Measurement Instruments

Flow meters are used to measure the flow of water, steam, fuel, and air throughout the plant. Maintaining proper flow rates is essential for efficient combustion and power generation.

Level Measurement Instruments

Level transmitters are used to monitor water levels in boiler drums, tanks, condensers, and storage vessels. Proper level control ensures safe operation and prevents equipment failures.

Distributed Control System (DCS)

The Distributed Control System (DCS) is the central control system used at NCTPS Stage-II. It continuously collects data from field instruments and displays important parameters in the control room. Operators use the DCS to monitor plant performance and control various processes.

The DCS automatically regulates critical parameters such as boiler pressure, steam temperature, feedwater flow, and turbine operation. It also generates alarms and warning signals whenever abnormal conditions occur. This centralized control system improves efficiency, reliability, and operational safety.

Control Valves and Actuators

Control valves are important final control elements used to regulate the flow of fluids in the plant. They receive signals from the control system and adjust their position accordingly. Pneumatic and electric actuators are commonly used to operate these valves.

Control valves play a major role in maintaining stable process conditions by regulating steam, water, fuel, and air flow throughout the power plant.

Maintenance and Calibration Activities

The C&I Department performs regular maintenance and calibration of instruments to ensure accurate measurements. Calibration involves comparing instrument readings with standard values and making necessary adjustments. Routine inspections, testing, troubleshooting, and preventive maintenance help maintain instrument reliability and reduce operational errors.

Proper calibration ensures accurate process monitoring and improves the overall performance of the plant.

Importance of Automation in Power Plants

Automation has become an essential part of modern thermal power plants. Automated control systems reduce human error, improve response time, and enable efficient operation of complex plant processes. The integration of instrumentation and control systems helps optimize fuel consumption, improve equipment performance, and maintain stable power generation.

Automation also enhances safety by providing early fault detection and automatic protective actions during abnormal operating conditions.

6.MECHANICAL REPAIR AND TESTING (MRT)

Introduction

The Mechanical Repair and Testing (MRT) Department at North Chennai Thermal Power Station Stage-II (NCTPS-II) is responsible for the maintenance, repair, testing, and refurbishment of mechanical equipment used throughout the power plant. The department plays a vital role in ensuring the reliable operation of boilers, turbines, pumps, valves, compressors, conveyors, and other auxiliary equipment. Proper maintenance and testing activities help prevent unexpected breakdowns, improve equipment efficiency, and extend the service life of plant machinery. The MRT department supports both preventive and corrective maintenance activities, thereby contributing significantly to continuous power generation.

Role of the MRT Department

The primary function of the MRT Department is to inspect, repair, and maintain mechanical equipment to ensure smooth plant operation. The department handles routine maintenance work as well as emergency repairs during equipment failures. MRT personnel work closely with other maintenance departments to identify mechanical issues and implement corrective actions.

The department also performs testing and condition monitoring of critical equipment to ensure that all machinery operates within safe and efficient limits. Through systematic maintenance practices, MRT helps improve plant reliability and operational performance.

Mechanical Workshop Activities

The MRT Department operates a well-equipped mechanical workshop where various repair and fabrication activities are carried out. The workshop contains machines and tools used for machining, grinding, cutting, drilling, and fabrication work.

Mechanical components such as shafts, bearings, couplings, bushings, and brackets are repaired or manufactured in the workshop whenever required. These activities help reduce downtime and ensure the availability of spare parts for maintenance work.

Welding and Fabrication

Welding and fabrication are important functions of the MRT Department. During maintenance activities, damaged structures, pipelines, supports, and equipment components are repaired using welding techniques. Various welding methods are employed depending on the type of material and application.

Fabrication work is carried out for the preparation of replacement parts, structural supports, and maintenance fixtures. Proper welding and fabrication ensure the structural integrity and reliability of plant equipment.

Pump and Valve Maintenance

Numerous pumps and valves are installed throughout NCTPS Stage-II for handling water, steam, oil, and other fluids. The MRT Department performs regular inspection, maintenance, and repair of these components.

Maintenance activities include checking seals, bearings, impellers, valve seats, and actuating mechanisms. Damaged components are repaired or replaced to ensure efficient fluid flow and prevent leakage. Proper maintenance of pumps and valves contributes significantly to plant efficiency and operational safety.

Testing and Inspection Activities

Testing and inspection form an essential part of MRT operations. Various mechanical components are inspected periodically to identify wear, corrosion, cracks, and other defects. Equipment condition is evaluated through visual inspection and performance testing.

Alignment checks, vibration monitoring, dimensional measurements, and operational testing are performed to ensure equipment reliability. These testing activities help identify potential problems at an early stage and reduce the risk of unexpected failures.

Preventive Maintenance Practices

The MRT Department follows preventive maintenance schedules to minimize equipment breakdowns. Routine inspections, lubrication, alignment verification, cleaning, and component replacement are carried out regularly.

Preventive maintenance helps improve equipment performance, reduce maintenance costs, and increase the operational life of machinery. Planned maintenance activities also reduce the likelihood of forced outages and improve plant availability.

Importance of MRT in Power Plant Operation

The Mechanical Repair and Testing Department plays a crucial role in maintaining the mechanical health of the power plant. Efficient operation of boilers, turbines, pumps, and auxiliary systems depends on the quality of maintenance performed by the MRT team. Their work ensures that equipment remains reliable, safe, and capable of supporting continuous electricity generation.

The department also contributes to reducing downtime, improving plant efficiency, and enhancing operational safety through systematic maintenance and testing practices.

7.Boiler Maintenance (BM)

Overview of boiler

The North Chennai Thermal Power Station Stage-II (NCTPS-II) operates with two large vertical water tube boilers, each supporting a 600 MW . These boilers handles the high pressure and high temperature to produce steam. The steam temperature is up to 540°C, and the boiler operates continuously to generate steam.

Construction

Each boiler stands approximately 80 meters long and 105 meters high with 1,178 water tubes arranged around the furnace walls. These tubes absorb the heat from coal combustion and converts the water into steam. These boilers are important to the power generation process, producing the high pressure steam needed to run turbines and generate electricity.

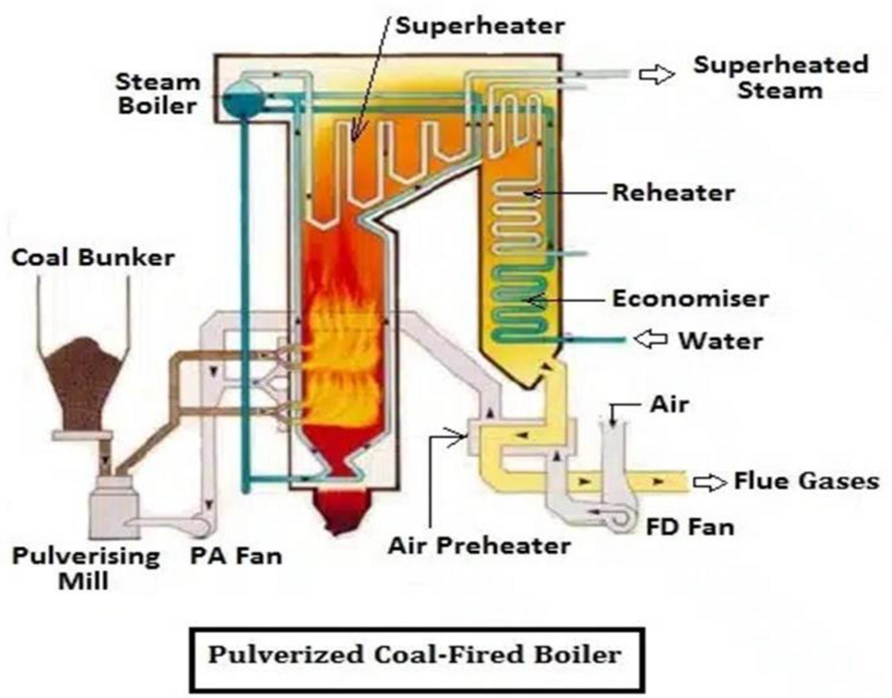

Fig 7.1 Boiler

Working of boiler

- The boiler consist of four corners and seven vertical elevations, that enables uniform heat distribution, combustion control and efficient air-fuel mixing .To start the boiler, the primary fuel ,pulverized bituminous coal is used and delivered into the furnace by Primary Air Fans (PAFs). Initially to fire the coal, Light Diesel Oil (LDO) and Heavy Furnace Oil (HFO) are used.
- The oil burners or oil guns, are mounted at the four corners of the furnace in a perpendicular layout, ensuring even flame spread and stable ignition.
- LDO is used to ignition the coal with temperature of 1500 C and HFO sustains combustion with temperature of 3000 C. Once a stable coal flame is established, the oil guns are withdrawn.

- To burn coal, a regulated and continuous air supply is essential. This is achieved by Forced Draft Fan (FDF), which draws the atmospheric air into the boiler by air preheater

Parts of the Boiler

1. Furnace

- This is where coal is burned with air to release heat.
- The coal is crushed into fine powder and sent into the furnace using primary air. Secondary air is also supplied for complete combustion.
- High temperatures (over 1300°C) are achieved here.

2. Boiler Drum

- A large cylindrical vessel that stores steam and water.
- It separates wet steam from water using separators and dryers.
- The water is circulated from the drum to water walls and back in a natural circulation loop.

3. Water Walls

- Tubes that lie the inside walls of the furnace.
- Water flows through these tubes and gets heated by radiation from the flame.
- It converts water into steam and moves up to the boiler drum.

4. Superheater

- Takes saturated steam from the drum and heats it further to become superheated steam with temperature of 540°C.

- Superheated steam is then sent to the High-Pressure (HP) Turbine. Multiple stages: platen superheater, pendant superheater, final superheater.

5. Reheater

- Steam that comes out of the HP turbine is reheated to high temperature again before entering the IP turbine.
- This improves efficiency and prevents water droplet formation in LP turbines.

6. Economizer

- Located in the flue gas path to absorb heat from the flue gas.

- Uses waste heat from flue gases to preheat the feedwater before it enters the boiler drum.
- Increases boiler efficiency.

7. Air Preheater (APH)

Uses hot flue gases to heat the incoming combustion air to give the required oxygen for the combustion of the coal.

Ash Handling from Boiler

Fly ash is collected from Electrostatic Precipitators (ESPs) connected to the flue gas outlet Bottom ash is collected in the hopper and removed by the conveyor system .

Ash is handled by hydraulic or pneumatic systems and sent to silos for disposal or reuse.

Scheduled & Capital Maintenance:

Annual minor maintenance, Each unit undergoes roughly 15 days of routine maintenance every year.

Involves deep inspection and cleaning of boilers, turbines, pumps, etc.

2. Boiler Punctures & Forced Outages

There have been recurring boiler punctures leading to forced shutdowns . Those incidents highlight the need for better monitoring of tube integrity and preventive measures.

3. Air-Preheater & Boiler Light-Up Protocols

Cleanliness checks & soot blower readiness before light-up. • Inspect oil burner valves; commission APH interlocks and inspect ports

Maintain correct atomization and oil/heater temperature (≈ 100 °C for HFO). Continuous soot-blowing during oil firing to prevent deposit build-up and fires.

- Early infrared fire detection and rapid response systems.

These steps help prevent air-preheater fires and ensure safe boiler ignition.

4. Water Chemistry & Feed-Water Treatment

- NCTPS uses demineralization to reduce hardness from ~1000 ppm to 0.1 ppm, preventing scale inside boilers.
- Purified boiler feedwater is essential for tube longevity and safety. This matches standard industry practices for high-pressure coal boilers.

5. Auxiliary Systems & Fly Ash Handling

- Economizer and ESP systems: Regular cleaning of fly ash from economizer, APH, ESP hoppers to prevent back-pressure and corrosion. Delays have caused forced outages and ESP faults.
- Ash-handling maintenance: Required system upkeep to prevent ash slurry leaks and environmental contamination. This makes maintenance smarter, faster, and helps avoid sudden breakdowns, keeping the power plant running smoothly.

8.TURBINE MAINTENANCE (TM)

Introduction

The Turbine Maintenance (TM) Department at North Chennai Thermal Power Station Stage-II (NCTPS-II) is responsible for the inspection, maintenance, testing, and repair of steam turbines used for power generation. The steam turbine is one of the most critical components of the power plant, as it converts the thermal energy of high-pressure steam into mechanical energy, which is then used to drive the generator. Proper maintenance of turbines is essential to ensure reliable operation, high efficiency, and uninterrupted power generation. The TM Department plays a significant role in preventing equipment failures, reducing downtime, and maintaining the overall performance of the generating units.

Steam Turbine System

At NCTPS Stage-II, high-pressure and high-temperature steam generated in the boiler is supplied to the turbine. The turbine consists of three main sections: High Pressure (HP) Turbine, Intermediate Pressure (IP) Turbine, and Low Pressure (LP) Turbine. Steam first enters the HP turbine, where a portion of its energy is extracted. The steam is then reheated in the boiler and sent to the IP turbine before finally expanding through the LP turbine.

The turbine shaft rotates at approximately 3000 RPM and is directly coupled to the generator for electricity production. Efficient operation of the turbine is essential for achieving maximum power generation and plant efficiency.

Turbine Inspection Activities

The TM Department carries out regular inspections to assess the condition of turbine components. During inspections, critical parts such as turbine blades, rotors, casings, bearings, seals, and couplings are examined for wear, corrosion, erosion, and mechanical damage.

Periodic inspections help identify potential problems before they lead to major failures. Detailed inspection records are maintained to monitor equipment condition and plan maintenance activities effectively.

Turbine Blade Maintenance

Turbine blades are among the most important components of a steam turbine. They are continuously exposed to high-pressure steam and rotating forces. Over time, blades may experience erosion, corrosion, or deposits that can affect turbine efficiency.

The TM Department performs regular cleaning, inspection, and repair of turbine blades. Damaged blades are repaired or replaced when necessary. Maintaining blade condition helps improve turbine performance and ensures smooth operation.

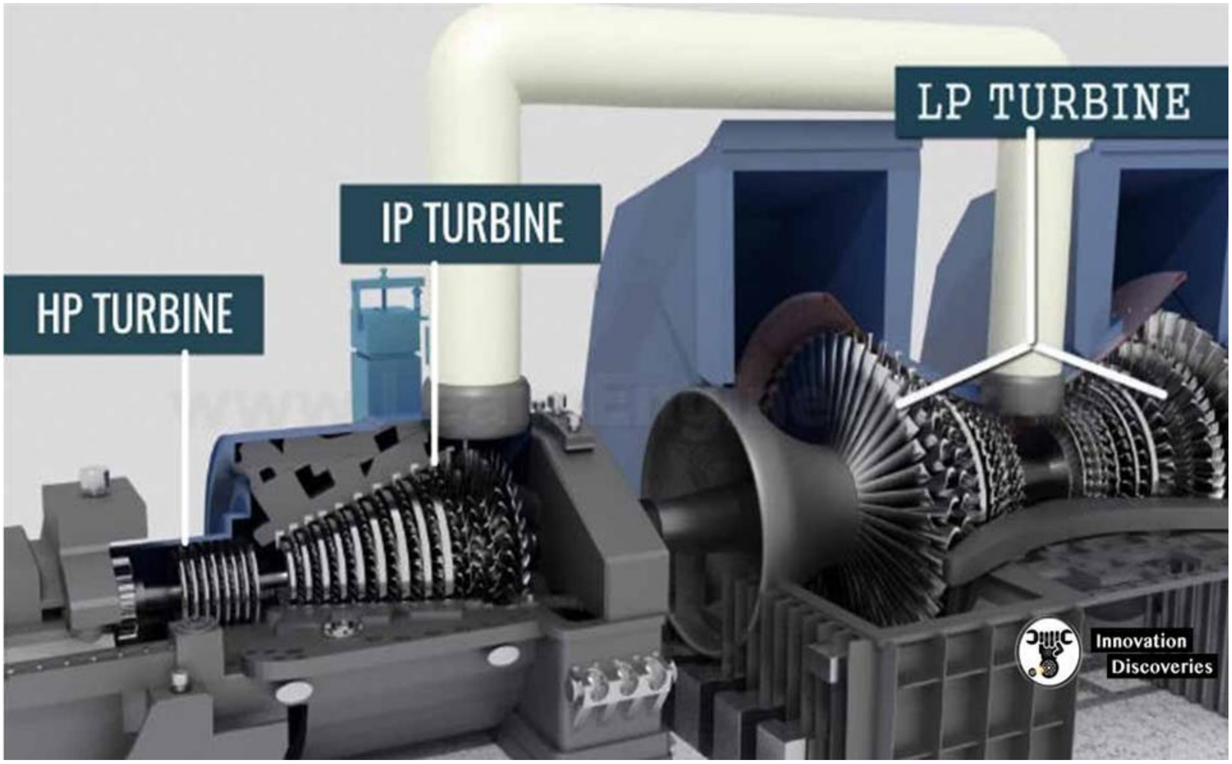

Fig 8.1 Turbine

Bearing and Lubrication System Maintenance

Bearings support the turbine rotor and allow smooth rotation at high speeds. Proper lubrication is essential to reduce friction and prevent excessive wear. The lubrication system supplies clean oil to bearings and other moving components.

The TM Department monitors oil quality, oil pressure, and bearing temperatures regularly. Oil samples are tested to identify contamination and ensure proper lubrication. Regular maintenance of the lubrication system helps prevent bearing failures and increases equipment reliability.

Vibration Monitoring and Alignment

Vibration monitoring is an important maintenance activity carried out by the TM Department. Excessive vibration may indicate imbalance, misalignment, bearing wear, or mechanical defects within the turbine.

Special instruments are used to measure vibration levels and identify abnormalities. Alignment checks are also performed to ensure proper positioning of the turbine and generator shafts. Early detection of vibration issues helps prevent costly equipment damage and unplanned shutdowns.

Preventive and Overhaul Maintenance

The TM Department follows preventive maintenance schedules to maintain turbine reliability. Routine activities include cleaning, inspection, lubrication, testing, and replacement of worn components. During planned shutdowns, major overhauls are carried out to inspect internal components thoroughly and restore equipment performance.

Preventive maintenance helps reduce forced outages, improve operational efficiency, and extend the service life of turbine equipment.

9.MILL PLANT (MP)

Introduction

The Mill Plant (MP) Department at North Chennai Thermal Power Station Stage-II (NCTPS-II) is responsible for the preparation and supply of pulverized coal to the boiler for efficient combustion. Since coal is the primary fuel used for power generation, proper operation and maintenance of the mill plant are essential for continuous and reliable plant performance. The main function of the mill plant is to crush and pulverize raw coal into fine powder so that it can burn efficiently inside the boiler furnace. Efficient coal pulverization improves combustion efficiency, reduces fuel consumption, and enhances overall power plant performance.

Role of the Mill Plant

The Mill Plant serves as a vital link between the Coal Handling Plant and the Boiler System. Coal received from the coal handling system is transported to the bunkers and then fed into the mills. Inside the mill, coal is pulverized into fine particles and mixed with hot primary air before being supplied to the boiler burners.

The quality of pulverized coal directly affects boiler efficiency, steam generation, and power output. Therefore, proper maintenance and monitoring of the mill plant are necessary to ensure smooth operation of the power station.

Bowl Mill System

NCTPS Stage-II uses Bowl Mills for coal pulverization. The bowl mill is one of the most important components of the mill plant. It consists of a rotating bowl and grinding rollers that crush the coal into fine powder.

Coal enters the mill through the feeder and is ground between the rotating bowl and rollers. Hot air supplied from the Primary Air Fan dries the coal and carries the pulverized coal to the boiler burners. The fine coal particles are then burned efficiently inside the furnace to generate heat for steam production.

Coal Feeders

Coal feeders regulate the quantity of coal supplied to the bowl mills. These feeders ensure a continuous and uniform flow of coal based on the boiler load requirements.

The performance of coal feeders is continuously monitored to maintain proper fuel supply. Regular inspection and maintenance of feeders help prevent blockages, uneven feeding, and operational interruptions.

Primary Air System

The Primary Air (PA) System plays an important role in the mill plant operation. Hot air generated by the air preheater is supplied to the bowl mills through primary air fans.

The primary air performs two important functions:

- Drying the moisture present in coal.
- Transporting pulverized coal from the mill to the boiler burners.

Maintaining proper primary air flow is essential for efficient pulverization and combustion.

Maintenance Activities

The Mill Plant Department carries out regular inspection and maintenance of bowl mills, coal feeders, primary air ducts, and associated equipment. Maintenance activities include checking grinding rollers, mill liners, bearings, gearboxes, and lubrication systems.

Vibration monitoring, temperature measurement, and wear inspections are performed periodically to identify potential problems before they cause equipment failure. Preventive maintenance helps improve equipment reliability and reduces unplanned shutdowns.

Safety Measures

Safety is an important aspect of mill plant operation. Coal dust can be highly combustible and may create fire hazards if not handled properly. Therefore, proper housekeeping, dust control systems, fire detection equipment, and safety procedures are strictly followed.

Maintenance personnel use appropriate Personal Protective Equipment (PPE) and follow safety guidelines while performing inspection and repair activities.

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Fig 9.1 Mill Plant.

10.OPERATION AND EXECUTION DIVISION (O&D)

Introduction

The Operation and Execution Division (O&D) at North Chennai Thermal Power Station Stage-II (NCTPS-II) is responsible for monitoring and operating various plant systems required for continuous power generation. The department ensures that all equipment operates safely, efficiently, and according to operational requirements. During the internship, exposure was provided to important operational systems such as the Cooling Water System and the Electrostatic Precipitator (ESP), which play a crucial role in maintaining plant efficiency and environmental compliance.

Role of the Operation and Execution Division

The O&D Division continuously monitors plant parameters and coordinates the operation of different auxiliary systems. The department is responsible for maintaining uninterrupted plant operation, monitoring equipment performance, and ensuring that all systems function within prescribed limits.

Operators monitor various parameters such as temperature, pressure, flow rate, and water levels through control systems. Proper coordination between different departments helps achieve stable and efficient power generation.

Cooling Water System

The Cooling Water System is one of the most important auxiliary systems in NCTPS Stage-II. Large quantities of cooling water are required to condense the exhaust steam coming from the turbine. The station utilizes seawater as the cooling medium because of its availability and effectiveness in heat removal.

The cooling water is drawn from the sea through intake structures and pumped to the condenser. Inside the condenser, heat from the exhaust steam is transferred to the cooling water, causing the steam to condense back into water. The condensed water is then returned to the feedwater cycle for reuse in the boiler.

Efficient cooling water circulation helps maintain condenser vacuum and improves turbine efficiency. Continuous monitoring of pumps, pipelines, valves, and cooling water flow is necessary to ensure reliable plant operation.

Importance of Cooling Water System

The Cooling Water System plays a significant role in improving the efficiency of the thermal power plant. Proper cooling ensures effective condensation of steam, which increases turbine performance and reduces energy losses.

The system also helps protect equipment from overheating and supports continuous operation of the generating units. Regular inspection and maintenance of cooling water pumps and associated equipment are essential for maintaining operational reliability.

Electrostatic Precipitator (ESP)

The Electrostatic Precipitator (ESP) is an important pollution control equipment installed in NCTPS Stage-II. Since coal combustion produces large quantities of fly ash, the ESP is used to remove ash particles from flue gases before they are released into the atmosphere.

The ESP operates on the principle of electrostatic attraction. High-voltage electrodes create an electric field that charges the fly ash particles present in the flue gas. These charged particles are attracted towards collecting plates, where they accumulate and are periodically removed through rapping mechanisms.

The collected fly ash is transferred to the Ash Handling System for storage and disposal. This process helps minimize air pollution and ensures compliance with environmental regulations.

Importance of ESP

The Electrostatic Precipitator is essential for controlling particulate emissions from the power plant. It significantly reduces the amount of fly ash released into the atmosphere, thereby protecting the environment and public health.

Efficient operation of the ESP improves air quality and helps the plant meet pollution control standards established by environmental authorities. Regular monitoring and maintenance of ESP components are necessary to maintain high collection efficiency.

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